Hostel Management System

Setup & Demo Guide for Windows

Run the complete project on your own computer in about 20 minutes: React frontend, Spring Boot backend and MySQL database - with demo data included.

1. What you need

Install these four things first (all free). You only need MySQL from XAMPP - Apache web server is not required.

- Java JDK 17 or newer - check with: java -version
- Apache Maven 3.9+ - check with: mvn -version
- Node.js 18+ (includes npm) - check with: node -version
- XAMPP (for MySQL only) - we just press Start next to MySQL

Official download pages:

Java (Adoptium Temurin): https://adoptium.net/
Maven: https://maven.apache.org/download.cgi
Node.js (LTS): https://nodejs.org/
XAMPP: https://www.apachefriends.org/
Tip: install everything with default options, then close and reopen PowerShell before continuing.

2. Get the project folder

Pick ONE option:

Option A - copy from your friend (easiest)

- Copy the whole hostel-management-system folder from USB / Google Drive.
- Place it somewhere simple, e.g. C:\hms (short path, no special characters).
- Everything is included, including the backend/.env settings file.

Option B - clone from GitHub

```
git clone <your-github-repo-url>
cd hostel-management-system
copy backend\.env.example backend\.env
```

The copy step is required: backend/.env holds your private settings (database login, admin password) and is never stored in git.

3. Start MySQL (XAMPP)

- Open XAMPP Control Panel and click Start next to MySQL.
- It should turn green and show port 3306. That is all - no password setup needed.
- XAMPP uses user root with an EMPTY password, which already matches the backend defaults.

- If port 3306 is busy, stop the other MySQL first (only one MySQL can use 3306).

You do NOT need to create any database or tables. The backend creates the hostel_db database, all 7 tables and the demo data automatically on its first start.

4. Start the backend (Spring Boot)

Open a PowerShell terminal and run:

```
cd C:\hms\backend
mvn spring-boot:run
```

- First run downloads libraries from the internet (3-5 minutes). Later starts take ~15 seconds.
- Success looks like: Started HostelApplication in ... seconds
- You will also see Flyway lines (Migrating schema...) - that is the database being built.
- API runs at http://localhost:8080. Keep this terminal open while using the app.

If your MySQL has a different root password, set it in backend/.env (DB_USERNAME / DB_PASSWORD) and restart the backend.

5. Start the frontend (React)

Open a SECOND PowerShell terminal and run:

```
cd C:\hms\frontend
npm install   # only the very first time (1-2 minutes)
npm run dev
```

- Success looks like: Local: http://localhost:5173/
- Open that address in Chrome/Edge. Keep this terminal open too.
- To stop either program later, press Ctrl+C in its terminal.

6. Log in and demo

On http://localhost:5173/login enter the admin login from your backend/.env file (defaults shown):

```
Username:  admin
Password:  admin123
```

Each computer has its own separate copy of the data: anything you add, edit or delete stays on your machine only.

What to show (2 minutes per page):

- Dashboard - live totals, occupancy chart, fee collection chart, recent students, pending complaints.
- Students - search, Add Student, open details with the ... menu, edit, delete.
- Rooms - occupancy bars, add a room, try lowering capacity below occupancy (blocked).
- Room Allotment - New Allotment, try double-booking a student (blocked), Check Out a student.
- Fees & Payments - Record Payment, partial payment updates the balance, View Receipt, print it.
- Complaints - Add Complaint, advance Open -> In Progress -> Resolved with one click.

7. If something goes wrong

"mvn" is not recognized
Maven is not installed or not on PATH. Reinstall Maven, reopen PowerShell.

Backend DB / connection errors
MySQL is not running. Start it in XAMPP. Check DB settings in backend/.env.

Port 8080 / 5173 / 3306 in use
Close the program using that port (another MySQL, server, or Skype) and retry.

Login fails
Use ADMIN_USERNAME / ADMIN_PASSWORD from YOUR backend/.env, then restart the backend.

npm install fails
Needs internet. Delete frontend/node_modules and run npm install again.

Blank white page
Backend is probably stopped. Start it first, then refresh the browser.

Flyway checksum error (rare)
Your database came from an older version. Run DROP DATABASE hostel_db; in MySQL and restart the backend.

Useful addresses: frontend http://localhost:5173 | backend http://localhost:8080